Telephones

Telephones make conversation possible when people are too far apart to be heard directly. They convert sound waves into signals that can be rapidly transmitted to another telephone, where the speech is reproduced.

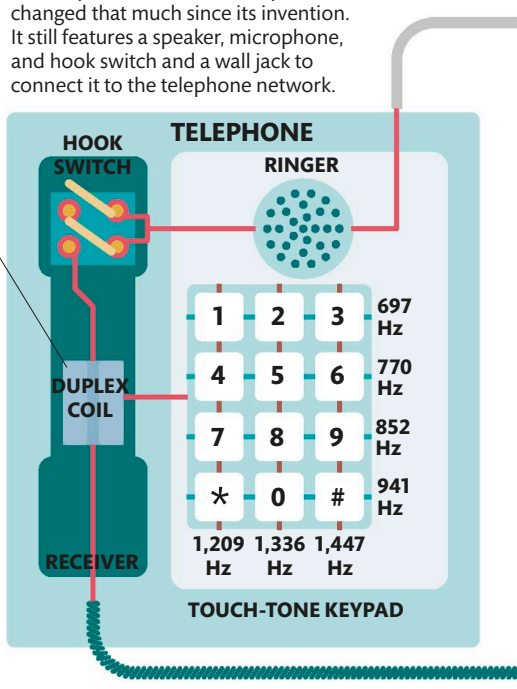
How telephones work

A person begins a call by picking up his or her handset and dialing a number to reach a recipient's telephone. Picking up the ringing telephone connects the speakers. The caller's speech travels through the telephone network in the form of electrical, optical, or radio signals, before being reproduced by the other telephone. Telephones contain both transmitters and receivers, which allow for two-way communication.

1 Connecting to an exchange
A switch called a hook switch connects and disconnects the telephone from the telephone network. Lifting up the telephone to make a call operates a lever, which forms an electrical connection between the handset and the local telephone exchange.

2 Dialing a number
Entering a digit on the keypad produces a distinct sound comprising two simultaneous frequencies, one high and one low. For example, key 7 produces a signal composed of tones with frequencies of 852 and 1,209 Hz. This unique sequence in a telephone number indicates to the exchange where the call should be directed to.

Telephone anatomy
Except for the development of keypads, the telephone's basic anatomy has not changed that much since its invention. It still features a speaker, microphone, and hook switch and a wall jack to connect it to the telephone network.



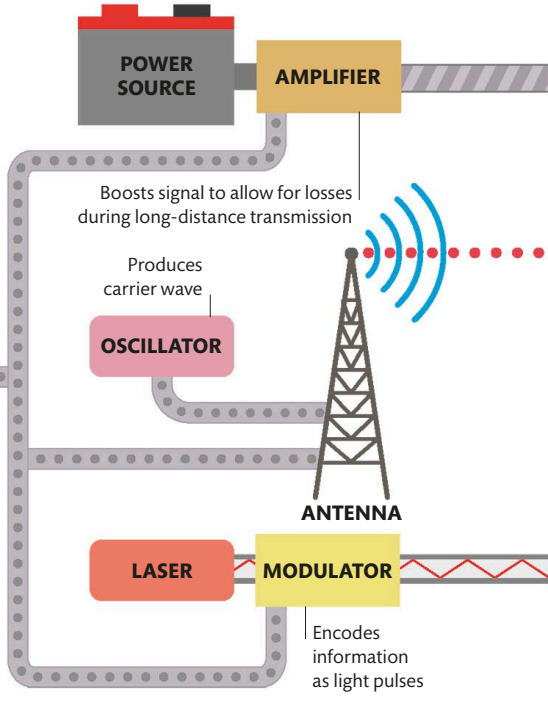
WHAT WERE THE FIRST WORDS SPOKEN ON THE TELEPHONE?

"Mr. Watson, come here; I want to see you," were the words uttered by telephone inventor Alexander Graham Bell on March 10, 1876, to his assistant.

Three transmission methods
Most information in the public switched telephone network is carried in the form of electrical, optical, or radio signals. These move much faster than the speed of sound.

1 Capturing a signal
A microphone in the mouthpiece converts sound waves into electrical signals of the same frequencies. These can travel through the telephone network in three different ways.





Delivery of sound

Telephone conversations sound natural when the signals travel quickly, with minimal delay. Sound waves are turned into electrical signals and travel through the telephone network as electrical or electromagnetic signals, before being transformed back into sound at their destination. This makes transmission so fast that it feels instantaneous, even on long-distance calls.